
CIS 6270 · PROJECT 1 DEFENSE · GROUP 2

BDG: Batch-Dispersion Guidance for Property-Band Generation

A feedback-servoed deviation gain for property-gradient guidance, studied on QM9 3D molecules and on DeepFlyBrain DNA enhancers on the probability simplex

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Frame the study

Problem. Steer a frozen generator so that each sample's property lands in a band $y \pm \delta$ around a target. The score is **band coverage**: the fraction of samples whose property falls inside the band.

Question. In gradient guidance, one strength w both moves the batch toward y and shrinks its spread. Does setting the spread separately, by feedback on the batch, raise coverage beyond tuning w ?

Hypothesis. Adjusting dispersion separately from centring raises band coverage at comparable sample quality, with the same rule in both state spaces.

MODALITY 1

QM9 3D molecules [1]

Coordinates in $\mathbb{R}^{N \times 3}$ plus atom types relaxed onto a simplex, $N \leq 29$. Symmetric under translation, rotation and permutation [2].

MODALITY 2

DeepFlyBrain DNA enhancers [4]

500-bp sequences as points of $(\Delta^3)^{500}$: one probability simplex over $\{A, C, G, T\}$ per position. No rotation group; every position must stay on the simplex.

Two state spaces, not two datasets: a Euclidean space acted on by a symmetry group, and a product of probability simplices.

Datasets, splits and leakage

MODALITY 1 · QM9 [1]

133,885 molecules, explicit hydrogens

MoleculeNet release, pinned by SHA-256. We keep the 3,054 molecules that EDM's preprocessing drops [2].

Split	Used for	n
train_a	generator and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	held out	13,083

No leakage. f_B, which scores every sample, never saw a molecule that trained the generator or the guide. Normalisation uses train_a alone.

Preprocessing. QM9: coordinates centred per molecule (Å), padded to 29 atoms with a mask; properties in D, Bohr³ and Ha. DNA: one-hot on the simplex, decoded by per-position argmax.

MODALITY 2 · DEEPFLYBRAIN [4]

Drosophila enhancers, 500 bp

One-hot A, C, G, T; four ambiguous training sequences removed.

Split	Used for	n
train	generator	83,722
val	checkpoint selection	10,505
test	held out	10,434

No learned evaluator. GC and CpG are counted exactly on the decoded sequence, so there is nothing for guidance to exploit and nothing to leak.

Cell-type labels are not given to the generator. A random split says nothing about unseen sequence families.

Representations, conditioning and handling

	QM9 molecules	DeepFlyBrain DNA
Representation	Each atom: 3D coordinates plus a one-hot type over {H, C, N, O, F}. Types are relaxed to continuous values during generation and read back by argmax.	Each position: a probability vector over {A, C, G, T}. A sequence is a point of $(\Delta^3)^{500}$, the simplex view of Dirichlet FM and Fisher Flow [4, 5]; read back by argmax.
Conditioning	None on the property. The only other input is the atom mask, drawn from QM9's size distribution and shared by every arm.	None. Cell-type labels are not given to the generator.
Symmetry and constraints	Translation: noise projected to zero centre of mass. Rotation, reflection, permutation: an E(3)-equivariant EGNN [9]; property nets are invariant. 12/12 equivariance checks pass.	No rotation group. After each step, values are clamped at zero and renormalised per position, so every position stays on the simplex.

Unconditional on purpose. The property enters only at sampling time, through guidance, which is exactly where BDG acts. Both modalities therefore test the same thing.

Flow-matching baseline: the formulation

Flow matching learns a velocity field v_θ that carries noise x_0 to data x_1 along a straight line [6, 7, 8]. Time t runs from 0 (noise) to 1 (data).

Path and target

$$x_t = t x_1 + (1 - t) x_0$$

$$u_t(x_t \mid x_0, x_1) = x_1 - x_0$$

The target is conditioned on the pair (x_0, x_1) , not on any property: v_θ has no condition input.

Noise source

Coordinate noise is projected to zero centre of mass (dimension $3(N - 1)$); type noise is a masked Gaussian.

Loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left\| v_\theta(x_t, t) - (x_1 - x_0) \right\|_M^2, \quad \|z\|_M^2 = \frac{\|M \odot z^c\|^2}{3 \sum M} + \frac{\|M \odot z^f\|^2}{5 \sum M}$$

M is the atom mask. Coordinate (c) and type (f) errors are averaged separately and weighted equally, so padding never dilutes the loss.

Endpoint estimate

$$\hat{x}_1 = x_t + (1 - t) v_\theta(x_t, t)$$

A one-step estimate of the final sample at any t . Guidance differentiates it (2b).

Flow matching: training, selection, sampling

Architecture [9]

E(3)-equivariant graph network (EGNN): width 256, 8 layers, SiLU, 3,753,229 parameters. Time enters through the input embedding; no attention.

Training

train_a only. Adam, learning rate 2×10^{-4} with cosine decay to 5%, batch 256, gradient clip 5, EMA 0.9999. 1,500 epochs (303k steps); seed 20260918.

Model selection

The training code keeps the best validation atom stability at NFE 100 (loss as tie-break), scored on one 512-sample draw every 25 epochs.

Disclosed. Every result in this talk uses the recorded final-epoch EMA weights (epoch 1,500; MD5 e19ccc06), not the rule's pick. Our diffusion model must report its evaluated checkpoint the same way (1d).

Sampling [6]

Probability-flow ODE, explicit Euler, 100 uniform steps, so **NFE 100**. Unguided, one sample takes 0.051 s on an RTX A6000, scoring included (2a).

NFE (number of function evaluations): how many times the solver evaluates the learned field to draw one sample, the standard cost unit for ODE samplers. Flow matching: 100; our diffusion model: 101 (1d). Guidance adds derivative calls on top of NFE; 2e counts them.

Diffusion baseline: the formulation

Variance-preserving (VP) diffusion in continuous time [2, 10, 11]. Noising time u runs from 0 (data) to 1 (noise); the network ε_ϕ predicts the added noise.

Noise schedule	$\beta(u) = \beta_{\min} + u (\beta_{\max} - \beta_{\min}) = 0.1 + 19.9 u$
Forward process	$x_u = a(u) x_1 + \sigma(u) \varepsilon, \quad a(u) = e^{-\frac{1}{2} (0.1 u + 9.95 u^2)}, \quad \sigma^2 = 1 - a^2$
Loss	$\mathcal{L}_{\text{VP}}(\phi) = \mathbb{E} \left\ \varepsilon_\phi(x_u, u) - \varepsilon \right\ _M^2$ The same masked norm as flow matching; ε projected to zero centre of mass; $u \sim \text{U}(10^{-3}, 1)$ keeps σ away from zero.
Sampler	$\frac{dx}{du} = -\frac{1}{2} \beta(u) (x + s_\phi(x, u)), \quad s_\phi = -\varepsilon_\phi / \sigma$ Integrated from $u = 1$ to 0 on a decreasing 100-step grid, then one denoising step to $\hat{x}_1$: NFE 101.
Endpoint estimate	$\hat{x}_1 = (x_u - \sigma(u) \varepsilon_\phi(x_u, u)) / a(u)$ Tweedie's posterior mean: the diffusion counterpart of $x_t + (1 - t)v\theta$, and what guidance differentiates.

Checked: $a^2 + \sigma^2 = 1$ to 6 decimals, $a(0) = 1$, $a(1) = 0.0066$.

Diffusion: training, selection, sampling

Matched by design. Same representation, split (train_a), EGNN 256×8 , optimiser, learning-rate schedule, batch, EMA, clip, seed and 1,500-epoch budget as flow matching. Only the corruption process and the output head differ.

Architecture [9]

EGNN 256×8 , 3,753,229 parameters, with a noise (ϵ) head in place of the velocity head.

Model selection

The same code rule: best validation atom stability, loss as tie-break. The benchmark must state which checkpoint it evaluates; flow matching uses its final-epoch EMA (1c).

Sampling [2, 11]

100-step decreasing grid plus the final denoising step: NFE 101 against flow matching's 100. We report the extra evaluation rather than hide it.

Status

Training is done. Before any cell runs, the checkpoint is checked against the protocol: split, width and depth are enforced by the loader; epochs, batch, EMA and seed are reported. Then 18 cells: unguided and plug-in $\times \mu, \alpha$, gap $\times 3$ seeds $\times 2,000$, $w = 1$ **[TODO: v3 Job A]**. Until they land we claim no family winner (2a).

How we will measure the two models

Both base models are ours, and one held-out evaluator scores everything: **f_B**, trained on train_b, with the same band and the same metric code for every model and arm.

Control: band coverage (in-band)

Fraction of decoded samples with $|f_B(x) - y| \leq \delta$. Target y : the property's training median (q50). Band $\delta = 2 \times f_B$'s calibration MAE, fixed before any run.

Fidelity

Molecule stability (every atom's valence satisfied, EDM's rule [2]) and RDKit validity.

Diversity

Uniqueness among valid molecules, and distinct-valid yield (DV): distinct valid molecules per attempt.

Spread

R = standard deviation of f_B over decoded samples $\div$ the unguided value. A diagnostic of what guidance does to the batch, not a diversity measure.

Sample counts. Every cell, guided or not: 3 seeds (20261001–03) $\times$ 2,000 samples, drawn as 4 batches of 500, so each arm pools 6,000. The batch of 500 is also BDG's unit (2c).

What in-band cannot tell us. It inherits f_B 's error, and a guided arm can raise it by exploiting f_B , which is why f_A and f_B train on disjoint halves. It also counts invalid molecules, so it never appears without stability and validity beside it.

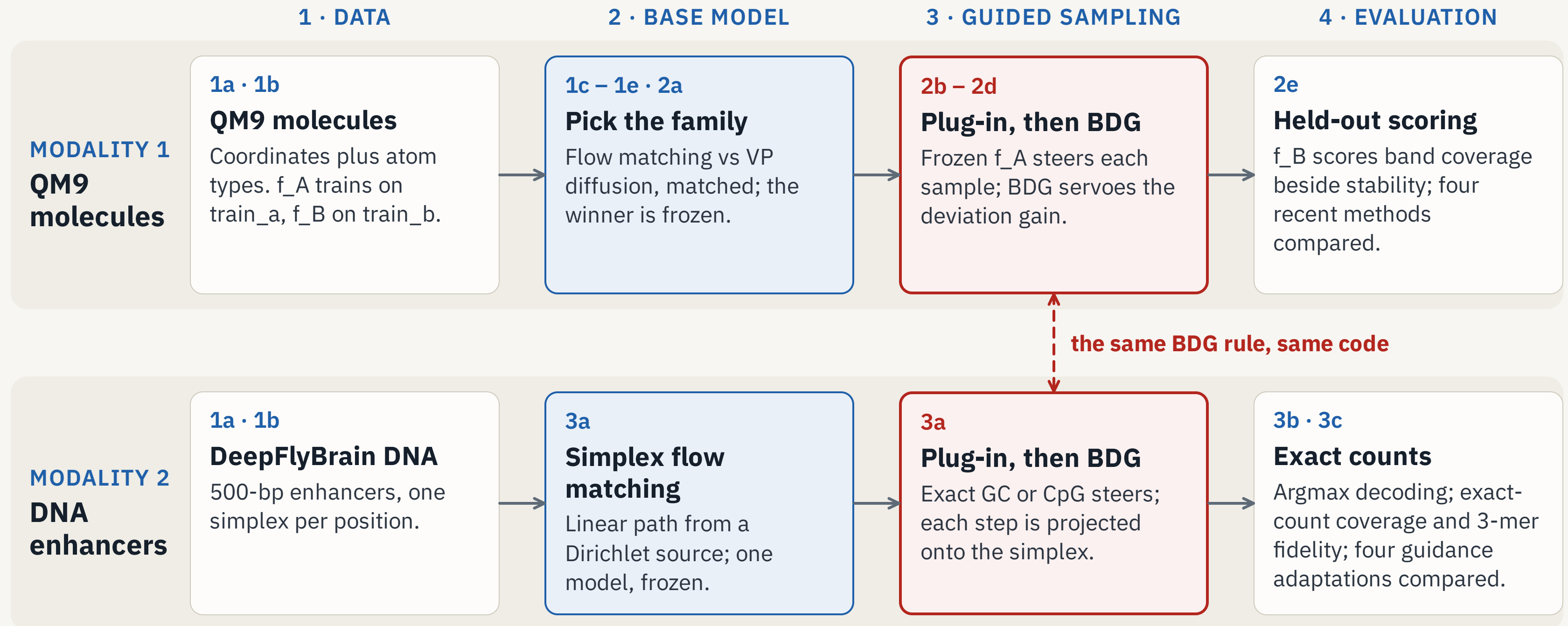
Flow matching vs diffusion: a matched protocol

Both models are ours, trained on the same half of QM9. Everything except the corruption process is held fixed, so 2a compares two families, not two engineering efforts.

Factor	Identical by construction	Necessarily different
Data and split	train_a, 51,527 molecules; same representation	—
Backbone	EGNN 256 × 8, 3.75M parameters each [9]	velocity head vs noise (ϵ) head
Corruption	—	linear interpolant vs variance-preserving: the variable under test
Training recipe	Adam 2×10^{-4} cosine, batch 256, clip 5, EMA 0.9999, seed 20260918, 1,500 epochs	VP: trained; recipe audit before its cells run [TODO: v3 Job A]
Checkpoint	—	FM: epoch-1,500 EMA; VP: stated with its benchmark
Sampling	probability-flow ODE, 100 steps	VP adds a denoising step: NFE 101 vs 100
Evaluation	same f_A , f_B , δ , atom masks, seeds, $3 \times 2,000$ per cell and metric code	—

Tested conservatively. Arms share noise, seeds and atom counts, yet the registered test is unpaired: binomial z on the pooled 6,000 per arm, bar 2.99 (18 contrasts, family level 0.05). Differences below the bar (about 0.8–1.9 points here) are reported as unresolved, never as equal.

Study overview: one guidance rule, two state spaces



■ trained by us ■ our change ■ data and scoring

Borrowed checkpoints (EquiFM, TFG's predictor pair) are separate comparisons, never pooled with ours.

Choose the base model

Every row is unconditional and unguided: no target, no property gradient. Guidance starts in 2b.

Model	Trained on	Atom ↑	Molecule ↑	Validity ↑	Unique ↑	NFE	s / sample
Flow matching (ours)	train_a (51,527)	93.6	39.7 ± 0.6	75.6 ± 1.1	99.8	100	0.051
VP diffusion (ours)	train_a (51,527)	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]	101	[TODO]
EquiFM [24], borrowed	full QM9 (100K)	98.6	86.2 ± 0.3	93.8 ± 0.6	99.7	100	0.086
Real QM9: the ceiling	—	99.4	95.6	98.2	—	—	—

Stability, validity and uniqueness in percent; mean ± seed s.d. over 3 seeds × 2,000 (atom and uniqueness pooled). Euler probability-flow ODE, 100 steps; our evaluator with EDM's conventions [2]; RTX A6000 time includes scoring. Real QM9: the same evaluator run on the data.

Flow matching goes forward, provisionally

It gives the guide an explicit endpoint, $m = x_t + (1 - t)v\theta$. The matched diffusion row is still pending, and EquiFM cannot stand in: an external model, twice our data, a different path.

The deciding test is **guided, not unguided**: 2b states it, and the choice stands only if flow matching passes.

Why we sit below the ceiling

- 1 · Unscaled one-hot features:** the published ablation removing exactly that scaling costs 3 atom and 35 molecule points [2].
- 2 · Training budget:** 303k steps against ~1.56M.
- Ruled out:** the half-data split (~0.3 points [25]) and our evaluator.

Guidance strategy: the plug-in rule

The rule: DPS-style plug-in [12]

$$G_i = \frac{y - F_i}{s^2} J_i^\top \nabla_m f_A(m_i), \quad F_i = f_A(m_i), \quad J_i = \frac{\partial m_i}{\partial x^{(i)}}$$

m_i is the model's one-step estimate of the final sample: $x_t + (1 - t)v_\theta$ for flow matching, the Tweedie mean for diffusion (1c, 1d). y is the property's training median (q50); s is its standard deviation over the 3,000-molecule calibration pool.

How it enters each sampler

$$\text{flow: } \frac{dx}{dt} = v_\theta + \text{clip}_{\|v_\theta\|} \left(w \frac{1-t}{t} G_i \right)$$

$$\text{diffusion: } \frac{dx}{du} = f_\phi + \text{clip}_{\|f_\phi\|} \left(-\frac{1}{2} \beta(u) w G_i \right), \quad f_\phi = -\frac{1}{2} \beta(u) (x + s_\phi)$$

The same G_i , strength $w = 1$ [14], cap and window in both: guidance acts only in the second half of sampling (flow $t \geq 0.5$, diffusion $u \leq 0.5$).

Frozen at sampling

Generator: ours, frozen.

Guide f_A : ours, trained on train_a; differentiated, never updated.

Evaluator f_B : trained on train_b; never inside the sampler.

Why the cap

Early in sampling $(1 - t)/t$ is large, and an uncapped correction can overwhelm the model's own velocity: trajectories diverge and coordinates overflow to NaN. In one diagnostic run, **222 molecules broke without the cap and none with it**. The cap is prior practice [13], identical for every arm.

Guided vs unguided, on both base models

Setting. QM9; target $y = q50$; the plug-in rule [12] at $w = 1$ for $t \geq 0.5$; f_A steers, held-out f_B scores decoded molecules ($\delta = 2 \times f_B$ calibration MAE). Each arm: 3 seeds $\times$ 2,000 in batches of 500, sharing noise, seeds and atom counts. Registered test: unpaired z , bar 2.99.

Property	Base model	In-band, decoded $\uparrow$	Change (z)	Mol. stab. $\uparrow$	Validity $\uparrow$
μ	Flow matching	$7.8 \pm 0.1 \rightarrow 10.6 \pm 0.4$	+2.75 (5.22)	39.7 $\rightarrow$ 35.6	75.6 $\rightarrow$ 72.5
μ	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]
α	Flow matching	$5.0 \pm 0.3 \rightarrow 6.1 \pm 0.3$	+1.07 (2.56)	39.7 $\rightarrow$ 37.2	75.6 $\rightarrow$ 73.7
α	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]
gap	Flow matching	$10.3 \pm 1.0 \rightarrow 12.6 \pm 0.7$	+2.35 (4.04)	39.7 $\rightarrow$ 33.6	75.6 $\rightarrow$ 69.9
gap	VP diffusion	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]

Each cell: unguided $\rightarrow$ plug-in, percent (mean $\pm$ seed s.d.). Change: percentage points, registered z in brackets. Uniqueness stays at 99.8–99.9.

What the flow model shows

Plug-in raises coverage on μ (**+2.75 points, z 5.22**) and **gap (+2.35, z 4.04)**; α (+1.07, z 2.56) is unresolved. The price: molecule stability falls 2.5–6.2 points and validity 1.9–5.7.

Which property decides the base model

Fixed before the diffusion cells land: we lead with the property where plug-in's coverage gain differs most between the two families, with its chemistry cost beside it. Flow matching stays the base model only if it wins there.

Motivating the innovation

The limitation

$$w(y - F_i) = \underbrace{w(y - \bar{F})}_{\text{moves the batch mean}} - \underbrace{w(F_i - \bar{F})}_{\text{shrinks the spread}}$$

In plug-in guidance [12] one strength w scales both terms, so moving the batch toward y also shrinks its spread.

BDG's idea, and the hypothesis

Keep the centring term; set the deviation coefficient by feedback on the batch variance V_b against a requested τ^2 . **Hypothesis:** coverage rises at comparable quality.

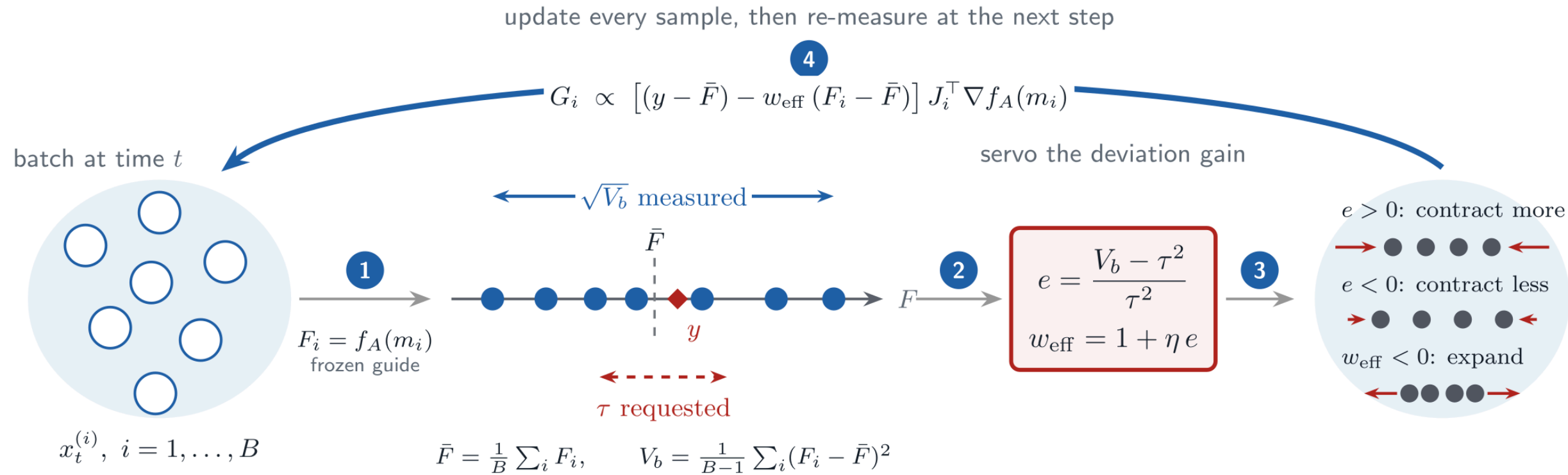
WHERE BDG SITS AMONG THE METHODS COMPARED IN 2E

Method	Statistic it uses	Acts on
DPS-style plug-in [12]	the residual $y - F_i$	one sample
TMPD-inspired [22]	+ the covariance of its own endpoint (a second moment)	one sample
LGD-MC [23]	Monte Carlo draws around the endpoint	one sample
TFG [15]	extra training-free steps on the endpoint	one sample
BDG (ours)	the batch variance of predicted properties, V_b vs τ^2 (a second moment)	the batch

Two different second moments. TMPD weighs one sample's uncertainty about its own endpoint; BDG reads the whole batch's spread in property space. No comparator uses a batch statistic.

Batch-level prior art, not re-run: joint potentials in Particle Guidance [20], moment targets in MGD [18], spread targets in VTD [19], the scale as a control gain in CFG-Ctrl and Feedback Guidance [16, 17], negative-guidance widening [21]. We claim only the composition (bounded search, ~8 queries).

BDG: the mechanism



BDG

$$n_i = \underbrace{(y - \bar{F})}_{\text{centring: fixed at 1}} - \underbrace{w_{\text{eff}}(F_i - \bar{F})}_{\text{deviation: } w_{\text{eff}}, \text{ servoed}}$$

no extra network calls
 $\eta = 0$: plug-in exactly
knobs: $\eta, \tau = \tau_{\text{mult}} s$
both terms scaled by w

One coefficient changes

The centring term keeps coefficient 1; $\eta = 0$ gives plug-in [12] exactly.

Same network calls

Both terms share one gradient; BDG adds two batch reductions.

Two knobs

Gain $\eta \geq 0$ and setpoint $\tau = \tau_{\text{m}} \cdot s$ (s : calibration-pool s.d., as in 2b).

BDG: the reduction and its limits

Exact reduction

$$\begin{aligned} n_i &= (y - F_i) - \eta e (F_i - \bar{F}) = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}) \\ &= w_{\text{eff}} (y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{w_{\text{eff}}} \quad (w_{\text{eff}} \neq 0) \end{aligned}$$

At any instant BDG is the plug-in direction with a batch-dependent gain w_{eff} and target y_{eff} . It adds no new direction.

When it contracts, and when it expands

$e > 0$ strengthens contraction; $e < 0$ only weakens it. Net expansion needs a negative gain:

$$w_{\text{eff}} < 0 \iff \eta > 1 \text{ and } V_b < \tau^2 (1 - 1/\eta)$$

A bound for any batch

$$V_b \geq 0 \implies e \geq -1 \implies w_{\text{eff}} \geq 1 - \eta$$

So BDG runs two-sided with no clamp.

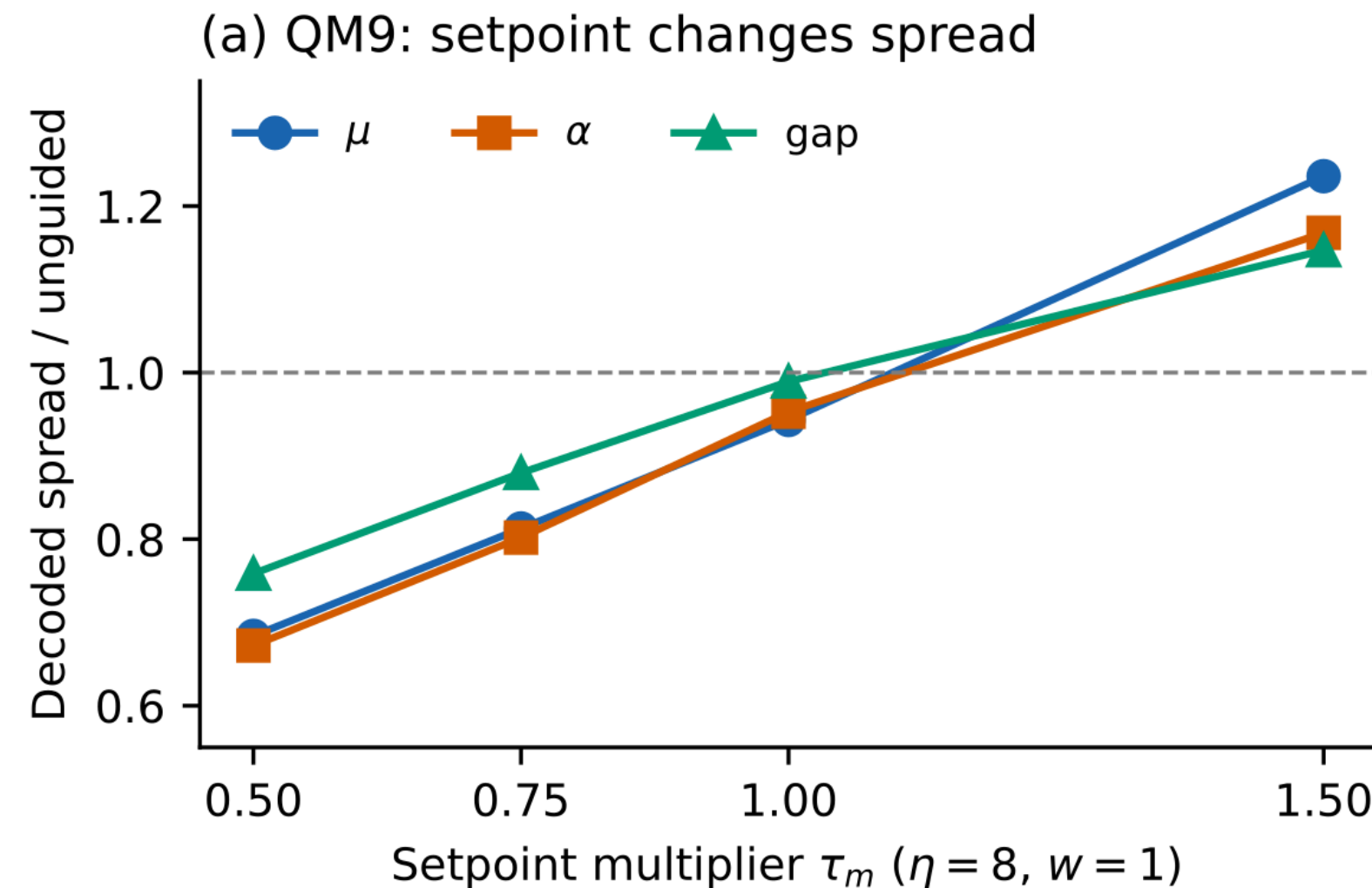
An idealised picture, not a guarantee

$$\dot{F}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}) \implies \dot{V}_b = -2 w_{\text{eff}} V_b \implies V^* = \tau^2 (1 - 1/\eta) \quad (\eta > 1)$$

Assumes unit response, continuous time and no drift, cap or projection. The stationary variance sits below the setpoint, and nothing guarantees the sampler reaches it.

What we do not claim: independent control of the realised mean and variance. The Jacobian and the cap act on both terms.

BDG at work: the setpoint moves the spread



Our FM, $\eta = 8, w = 1$; $3 \times 2,000$ per point. Dashed line: unguided spread.

Spread follows the setpoint

Raising τ_m from 0.5 to 1.5 moves decoded spread from **0.67–0.76×** to **1.15–1.24×** unguided, monotonically over four setpoints. EquiFM: 1.18–1.25× at $\tau_m = 1.5$.

Plug-in strength only narrows

Plug-in ($\eta = 0$) holds spread at 0.78–0.87× unguided at $w = 1$ and 0.65–0.79× at $w = 4$, in all six tasks.

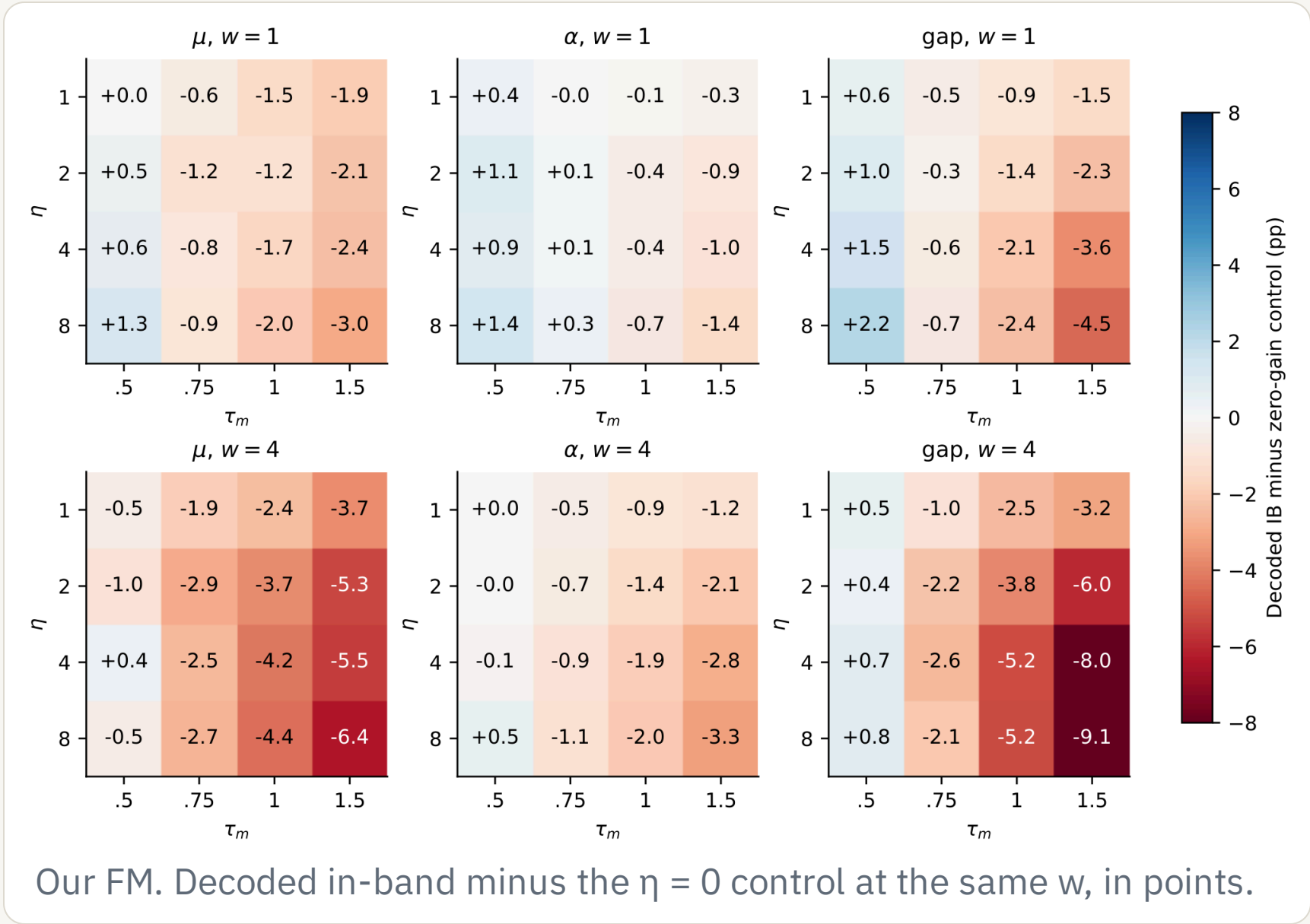
Same calls, exact limit

$\eta = 0$ reproduces plug-in bit for bit on our FM; EquiFM agrees within its rerun noise. BDG adds scalar reductions, no network call.

Shown: a wider operating range at fixed nominal strength. **Not shown:** higher coverage than plug-in (2d, 2e), or that the feedback itself is needed (no open-loop replay was run).

Testing the innovation: the ablation grid

612 cells. $\eta \in \{1, 2, 4, 8\} \times \tau_m \in \{0.5, 0.75, 1, 1.5\}$ plus $\eta = 0$, at $w = 1$ and 4 , on our FM and on EquiFM; $3 \times 2,000$ each. Every cell is read against its own $\eta = 0$ control at the same w .



Control · comparison	What we saw
Remove residual $\eta = 0$ vs plug-in	Identical on our FM, bit for bit; EquiFM within its rerun noise
Change strength $\eta = 0, w = 1 \rightarrow 4$	Spread narrows further: R 0.78–0.87 $\rightarrow$ 0.65–0.79
Change setpoint $\eta = 8, w = 1, \tau_m 0.5 \rightarrow 1.5$	Narrowing turns into widening: R 0.67–0.77 $\rightarrow$ 1.15–1.25, in 6 of 6 tasks
Change gain $\tau_m = 1.5, \eta 0 \rightarrow 8$	Coverage lower in all 12 grids (our FM, $w = 1$: –3.0, –1.4, –4.5 points)
Not run one-sided, replay, signed strength	One-sided residual, open-loop replay of $w_{\text{eff}}(t)$, signed-strength plug-in: limitations (3d)

R: decoded property s.d. $\div$ unguided. Grid cells are single contrasts against $\eta = 0$, read descriptively, not as headline tests.

What the ablations establish

If we observe (fixed in advance)	We conclude	Observed
$\eta = 0$ differs from plug-in	an implementation fault	No: identical on our FM; EquiFM within rerun noise
$\Delta IB > 0$ on all three seeds at comparable stability, and plug-in at $w = 4$ does not match it	BDG raises coverage beyond strength tuning: the full claim	Not met. Registered settings: 0 of 12 above plug-in, 3 below, 9 unresolved
R follows τ_m over three or more setpoints, but $\Delta IB \approx 0$	BDG sets spread, not coverage: the narrower claim	Met. R rises with τ_m over four setpoints; widening in 6 of 6
open-loop replay matches the closed loop	a schedule, not feedback	Not run
signed-strength plug-in matches BDG	scalar reweighting suffices	Not run

Why spread need not become coverage: a wider batch fills a fixed band less, and heterogeneous gradients, the cap and decoding break the algebra's promise.

Our reading

The supported claim is the narrower one: the setpoint sets spread at fixed nominal strength, widening past unguided where every tested plug-in strength narrows. It does not raise coverage over plug-in at the registered bar, and without replay it does not isolate feedback.

Coverage per point of chemistry

In-band points gained per stability point lost, from $\eta = 0$ at $w = 1$. Our FM: dispersion ($\eta = 8, \tau_m = 0.5$) 0.52–0.81 vs strength ($w 1 \rightarrow 4$) 0.38–0.53. EquiFM: 0.06–0.39 vs 0.08–0.37. Descriptive ratios, not tests.

Position against recent work

Method	In-band μ $\uparrow$	In-band α $\uparrow$	In-band gap $\uparrow$	Mol. stab. μ / α / gap $\uparrow$	Passes: gen. / guide	s / sample
Unguided	7.8 ± 0.1	5.0 ± 0.3	10.3 ± 1.0	39.7 / 39.7 / 39.7	400 / 0	0.051
DPS-style plug-in [12]	10.6 ± 0.4	6.1 ± 0.3	12.6 ± 0.7	35.6 / 37.2 / 33.6	1,000 / 400	0.119
TMPD-inspired [22]	9.0 ± 0.6	6.1 ± 0.4	11.6 ± 0.9	38.4 / 38.9 / 37.3	1,200 / 400	0.255
LGD-MC [23]	9.6 ± 0.8	5.2 ± 0.4	10.8 ± 0.7	39.9 / 39.2 / 39.5	1,200 / 2,000	0.229
TFG [15]	28.8 ± 1.6	10.5 ± 0.3	27.5 ± 0.3	24.5 / 26.8 / 14.4	800 / 2,000	0.122
BDG, $\tau_m = 0.5$ (ours)	11.2 ± 1.1	7.0 ± 0.5	14.2 ± 0.8	35.1 / 36.4 / 30.0	1,000 / 400	0.119
BDG, $\tau_m = 1$ (ours)	8.9 ± 0.2	5.6 ± 0.5	10.6 ± 0.2	40.5 / 37.9 / 37.6	1,000 / 400	0.119

Our FM, QM9, q50, $w = 1$, $t \geq 0.5$. Decoded in-band and molecule stability in percent (mean $\pm$ seed s.d., $3 \times 2,000$). Passes per 2,000-molecule cell. Time: RTX A6000, μ cells, scoring included.

Why these four

All are recent inference-time guidance rules that steer a frozen generator with a frozen predictor. Running them on *our* checkpoint isolates **the guidance rule**, which is what our contribution changes.

Efficiency, measured

BDG costs exactly plug-in's calls: **1,000 generator and 400 guide passes** per cell, 0.119 s per sample. LGD-MC and TFG need 5 $\times$ the guide passes; TMPD and LGD-MC take about twice the time. TFG uses fewer generator passes (800).

Fair study, and fair discussion

Reproduced, not quoted

Every 2e row is **run by us** on one frozen generator, guide, evaluator and target, with $3 \times 2,000$ samples per cell. No published score is pooled with ours. EquiFM's rows use TFG's predictor pair, so they are never ranked against our FM's.

Three port caveats

Plug-in is a DPS-*style* Gaussian-energy field; the original differentiates the residual L2 norm [12].

TMPD is the local-linear scalar limit: a related-work comparison, not a faithful nonlinear port [22].

TFG runs at its published QM9 hyperparameters, but through our deterministic ODE sampler, and our unscaled one-hot features move $\sim 16\times$ less relative to coordinates than the released checkpoint's.

Never read these against TFG's published table [15].

Better, measured

Cost. Plug-in's calls and time (0.119 s per sample): a fifth of LGD-MC's and TFG's guide passes, about half of TMPD's and LGD-MC's time.

Comparable, measured

Coverage vs plug-in. $\tau_m = 0.5$ is positive on all six tasks but unresolved (+0.45 to +1.52 points, $z \leq 2.44$), with 0.5–3.6 points less stability on our FM.

Worse, measured

$\tau_m = 1$ falls below plug-in in 3 of 6 tasks (−1.68 to −2.07 points). TFG gains far more coverage (above plug-in in 5 of 6) but costs 8–27 stability points.

The transferred method

Unchanged: the rule. $n_i, \bar{F}, V_b, e, \tau = \tau_{\text{mult}} \cdot s, \eta$ and $w_{\text{eff}} = 1 + \eta e$ carry over as written: one scalar per sample times the plug-in step, so no covariance matrix or inverse is needed (the simplex covariance $\text{diag}(p) - pp^T$ is singular).

What changes	Modality 1 · QM9	Modality 2 · DeepFlyBrain
State	$\mathbb{R}^{N \times 3}$ plus a type simplex	$(\Delta^3)^{500}$
Generator	EGNN flow matching, 3.75M	dilated 1-D CNN, 128×10 layers, $\approx 1.02\text{M}$; linear path from a Dirichlet(1, 1, 1, 1) source [4]
Guide	learned f_A	soft GC fraction or CpG frequency (analytic)
Evaluator	learned f_B	exact GC or CpG count after argmax
Feasibility	centring and atom masks	clamp at zero, renormalise each position

Training

83,722 sequences; 1,500 epochs (492k updates), batch 256, learning rate 2×10^{-3} cosine to zero, no EMA; validation kept epoch 1,450 (loss 0.0634); 539.7 min on an RTX A6000.

Sampling

100-step Euler, clamp and renormalise after every step, argmax at the end; guidance for $t \geq 0.5$, batch 500, 3 seeds $\times$ 2,000 per cell, $w \in \{1, 4\}$, $\eta = 4$, $\tau_m \in \{0.5, 1\}$.

$$F_{\text{GC}}(p) = \frac{1}{L} \sum_{j=1}^L (p_{j,C} + p_{j,G}), \quad F_{\text{CpG}}(p) = \frac{1}{L-1} \sum_{j=1}^{L-1} p_{j,C} p_{j+1,G}$$

GC is affine in p and CpG is quadratic, so the pair tests whether BDG depends on the property's curvature.
Known gap: the guide reads an endpoint that can leave the simplex; the evaluator counts decoded bases.

Transfer results: DeepFlyBrain DNA

DeepFlyBrain, length 500; GC and CpG in-band reported separately at strengths $w = 1$ and 4 ; 3-mer JSD names the fidelity cost. One generator, window and band for every guided row.

Method	GC, $w=1$ $\uparrow$	GC, $w=4$ $\uparrow$	CpG, $w=1$ $\uparrow$	CpG, $w=4$ $\uparrow$	JSD $w=4$ $\downarrow$
Unguided	14.0 ± 1.0	14.0 ± 1.0	50.2 ± 1.6	50.2 ± 1.6	4.53 / 4.53
DPS-style plug-in [12]	14.0 ± 0.9	14.1 ± 1.0	50.5 ± 1.8	51.6 ± 1.4	4.61 / 4.63
TMPD / LGD-MC [22, 23]	14.0 ± 0.9	14.1 ± 1.0	50.5 ± 1.8	51.6 ± 1.4	4.61 / 4.63
TFG-MC [15]	14.1 ± 1.0	14.1 ± 1.0	50.4 ± 1.7	51.1 ± 1.7	4.59 / 4.58
BDG, $\tau_m = 0.5$ (ours)	14.5 ± 1.0	16.4 ± 1.2	55.3 ± 0.8	68.5 ± 1.2	5.93 / 7.73
BDG, $\tau_m = 1$ (ours)	14.0 ± 1.0	14.1 ± 1.0	50.2 ± 1.6	50.3 ± 1.5	4.54 / 4.52

Decoded in-band, percent (mean $\pm$ seed s.d., $3 \times 2,000$). JSD $\times 10^{-4}$, GC / CpG at $w = 4$; unguided JSD 4.53. Plug-in, TMPD and LGD-MC coincide to $\leq 4 \times 10^{-6}$ here (one effective comparator). BDG – plug: GC +0.42 ($w1$), +2.32 ($w4$); CpG +4.82 ($w1$), +16.83 ($w4$).

Fair study

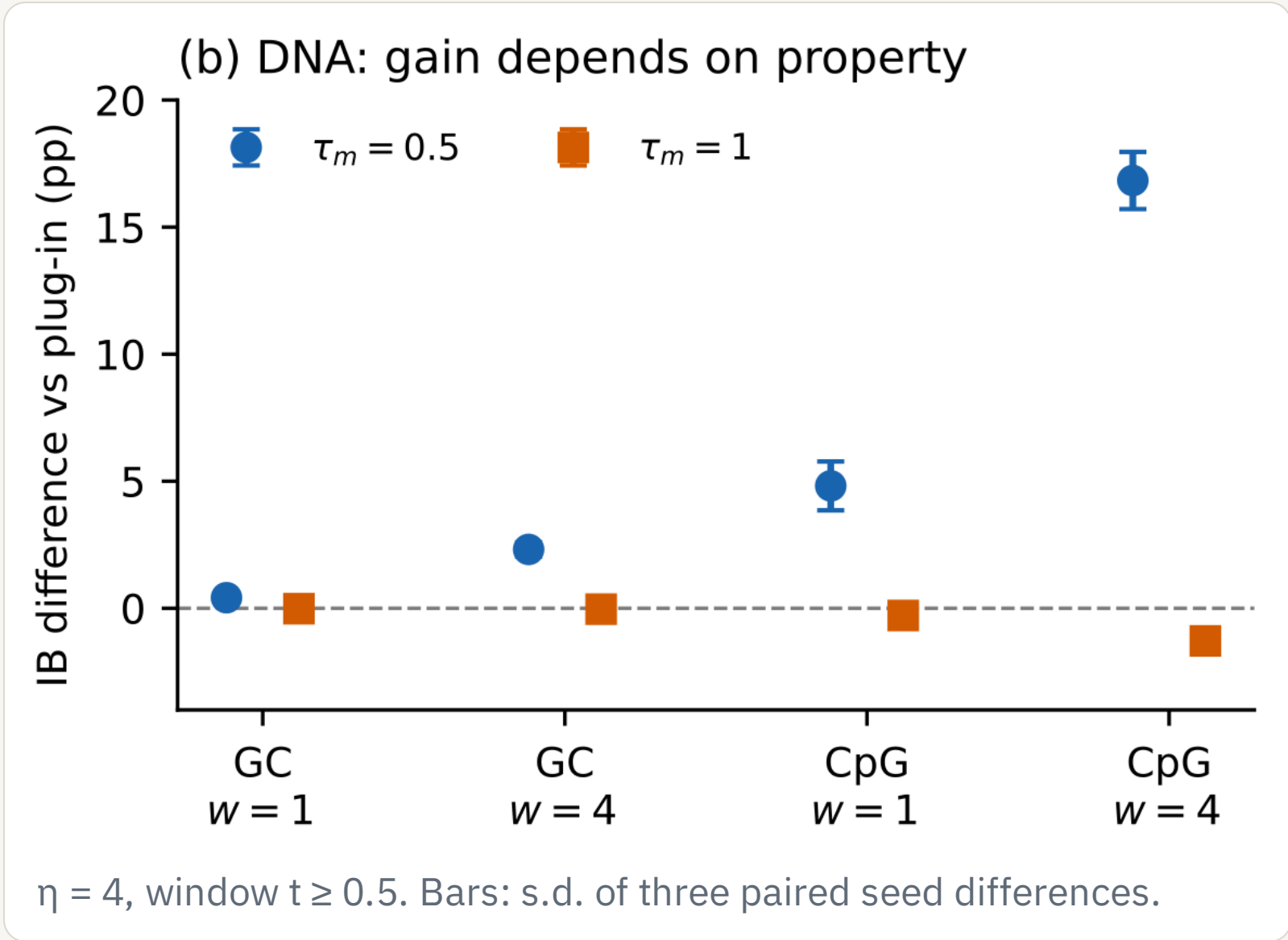
All rows share the generator, length, target, band, decoding and reference data. Endpoint uncertainty nearly vanishes late, so plug-in, TMPD and LGD-MC collapse onto one comparator; TFG-MC is a restricted port. Dirichlet FM, Fisher Flow and MOG-DFM are **not** evaluated here [3,4,5].

Fair discussion

The contracting setpoint ($\tau_m = 0.5$) raises CpG coverage at both strengths and GC at $w = 4$; $\tau_m = 1$ moves neither. The gains cost 3-mer fidelity (JSD up to 7.73 vs 4.53) and clip hard (CpG $w = 4$: 7.4% of steps).

Does the innovation transfer?

Criterion, fixed in advance. BDG transfers if its change from plug-in has the **same sign** in both modalities, in coverage and in how spread follows the setpoint, at a comparable fidelity cost. Each modality is judged against its own plug-in.



Modality	Property	Δ IB vs plug-in	Spread vs setpoint	Fidelity cost
QM9	μ, α, gap	0 / 12 above bar	follows τ_m (6/6)	-0.5 to -3.6 stab
DNA	GC	+0.4 (w1) · +2.3 (w4)	follows τ_m	JSD +0.4 to +1.4
DNA	CpG	+4.8 (w1) · +16.8 (w4)	follows τ_m	JSD +0.7 to +3.2

The verdict

Same direction, property-dependent size. On QM9 the coverage gain stays unresolved; on DNA the contracting setpoint clears the bar for **CpG at both strengths and GC at $w = 4$** . Spread follows the setpoint in both state spaces.

Why GC and CpG: GC is affine in the simplex coordinates, CpG quadratic. The gain is larger on the quadratic property; raw coverage is never compared across modalities, whose bands follow different rules.

Limitations and failure modes

1 · The evaluator bounds the claim

In-band inherits f_B 's error, and guidance can find its blind spots. Disjoint training halves reduce this; they do not remove it. DNA sidesteps it with exact counts.

2 · Guided $\neq$ scored

The guide reads a relaxed endpoint (continuous atom types, or DNA off the simplex) while scoring decodes. On GC at $w = 1$, early guidance added 22.8 points before decoding vs 4.2 in-window: a $\sim 5.4\times$ gap the decoded metric does not keep.

3 · Force, not feedback

Equal nominal w is not equal applied force: BDG's winning DNA cells clip 7.4–9.2% of steps against plug-in's $\leq 0.21\%$, and apply $\sim 10\times$ the correction share. A cap-limited change is not a controller effect.

4 · The batch is the estimator

$\bar{F}$ and V_b couple 500 samples, so uncertainty is over batches, not molecules; the registered z is unpaired and ignores this. EquiFM reruns add up to 0.7 points of noise the binomial se misses.

Also open, and reported as such: our VP base model is trained but not yet benchmarked (2a); no useful-yield (in-band *and* stable, paired) from the current aggregates; and the one-sided, open-loop-replay and signed-strength controls were not run, so feedback is not isolated from added force.

Conclusion and future directions

Does setting dispersion separately from centring raise band coverage at comparable quality?

**No on QM9.
Yes on CpG.**

No rung beats plug-in at the registered molecular bar; the contracting setpoint clears it for DNA CpG at both strengths and GC at $w = 4$.

What the evidence supports

BDG changes **one coefficient**, the deviation gain, by feedback on the batch variance; algebraically it is **plug-in with a batch-dependent gain and target**.

The setpoint moves decoded spread **past unguided in 6 of 6 tasks**, where every tested plug-in strength only narrows it.

The rule **transfers** across two state spaces at **plug-in's network calls**.

What would sharpen or withdraw it

Open-loop replay of $w_{\text{eff}}(t)$ matching the closed loop would show a schedule, not feedback; signed-strength plug-in matching BDG at equal force would show reweighting suffices. A matched-force comparison and batch-aware paired intervals would firm up the DNA gains.

Next experiments

- 1 • Benchmark our trained VP model: the 18-cell guided FM-vs-VP test (Job A).
- 2 • The not-run controls and paired, batch-aware useful-yield intervals.
- 3 • External sequence generators and a CpG ablation on DeepFlyBrain.

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APPENDIX · NOT PART OF THE 15-MINUTE TALK

Q&A backup

Five topics, one or two slides each.

A1

EquiFM, the borrowed generator

Architecture, path, and how our arms run on it

A2

Property predictors

Ours vs TFG's: architecture, training, accuracy

A3

The data split

Who trains on which half, and why halves

A4

The guidance arms

Each arm's correction, what it changes, what it costs

A5

Algorithm 1

One BDG-guided Euler step, line by line

EquiFM: the borrowed generator

Released checkpoint	
Paper	Song et al., NeurIPS 2023 [24]
Backbone	EDM-style equivariant GNN dynamics [2, 9], 256×9
Parameters	5,339,920 (EMA)
State per atom	3 coordinates (Å) + 5 one-hot ÷ 4 + charge ÷ 10; explicit H
Training data	the full 100K QM9 partition, unconditional

Hybrid probability path

Native time τ : 1 = noise, 0 = data.

Coordinates: almost linear — but the noise is rotated (Kabsch) and permuted (Hungarian) onto the molecule first, an equivariant optimal-transport coupling, plus an angle penalty.

Types and charge: variance-preserving, β from 0.1 to 20.

Sampling: Euler, $\tau = 1 \rightarrow 0$.

$$x_\tau = (1 - \tau) x_0 + s_x(\tau) z, \quad s_x = \epsilon + (1 - \epsilon)\tau$$

$$h_\tau = a h_0 + \sqrt{1 - a^2} z', \quad a = e^{-T(\tau)/2}$$

Unguided, same protocol as ours

3 seeds $\times$ 2,000, 100 Euler steps: atom **98.6%**, molecule **86.2 $\pm$ 0.3%**, validity **93.8 $\pm$ 0.6%**, uniqueness 99.7%; 0.086 s per sample on an RTX A6000.

Why its chemistry beats ours

Four candidates, none isolated by us: $\sim 2\times$ the training data, 5.34M vs 3.75M parameters, one-hot scaled by $\frac{1}{4}$ (our leading suspect in 2a), and the aligned coordinate path.

Running our guidance arms on EquiFM

Why it is in the study: a second, external flow-matching generator asks whether the comparison survives on a model we did not train. Backend **equifm**, all seven arms [24].

Endpoint estimate m	$m_x = (1 - \epsilon) x_\tau - s_x q_x, \quad m_h = a h_\tau + q_h/a$	Exact — algebra from its own regression targets.
Covariance Σ TMPD and LGD-MC only	$\Sigma = KJ, \quad k_x = \frac{s_x^2}{1 - \tau}, \quad k_h = \frac{1 - a^2}{a}$	Exact for types, approximate for coordinates , whose aligned noise is not an independent Gaussian channel.
Guided velocity	$\Delta v_x = -\frac{s_x}{1 - \tau} G_x, \quad \Delta v_h \approx -\frac{\beta(\tau)}{2} G_h$	The model's own velocity map at the guided mean $m + K \cdot G$.

Property pair here

TFG's released guide (tf_predict at time 0) and oracle (evaluate) [15]. Its band is narrower than ours — δ 0.157 vs 0.175 (μ), 0.165 vs 0.506 (α), 0.0037 vs 0.0075 (gap) — so **in-band never crosses backends**.

Disclosed limits

Trained on the full 100K, so it saw both halves. Only TMPD and LGD-MC touch the Σ surrogate; plug-in, TFG and BDG do not. The charge channel is held fixed during guidance. Same-seed reruns move coverage by up to 0.7 points; about 1.8× our peak memory at batch 500.

Property predictors: our pair and TFG's

Ours score the **fm** and **EDMsecond** backends; TFG's score **EquiFM**. f_A's **gradient** steers sampling; f_B only **scores**.

	Ours — f_A and f_B	TFG guide (tf_predict)	TFG oracle (evaluate)
Network	invariant EGNN on squared distances, mean-pool, linear head [9]	EDM dynamics EGNN with an energy head [2, 15]	EDM property-classifier EGNN [2]
Width × layers	128 × 4	192 × 7	128 × 7
Edge attention	no — equal-weight sum	yes	yes
Parameters (μ)	429,825 each	1,671,566	743,944
Inputs	coordinates + unscaled one-hot	+ one-hot ÷ 8; time fixed at 0	+ nuclear-charge powers
Training data	f_A: train_a · f_B: train_b (51,527 each)	QM9 second half (~50K)	QM9 first half (~50K)
Recipe	clean molecules, MSE on a standardised target; Adam 3e-4 cosine, batch 128, 120 epochs (~23 min); best validation MAE	clean molecules; EMA at epoch 2,000	best-validation checkpoint

Theirs are better; ours are disjoint

MAE	μ (D)	α (Bohr ³)	gap (Ha)
our f_A	0.0872	0.2325	0.00402
our f_B	0.0877	0.2531	0.00374
TFG guide	0.0417	0.0925	0.00185
TFG oracle	0.0784	0.0824	0.00187

Same 3,000 calibration molecules. Ours are scored only on the half each did not train on; TFG's on the whole pool, which overlaps their training halves and flatters them.

The honest gap on μ . On molecules neither TFG net trained on: guide 0.0659 D, oracle 0.0728 D, our f_B 0.0840 D. **~20% better, not 2×.**

Why theirs are more accurate
Capacity: 2–4× the parameters, 7 layers not 4.
Attention: a learned weight per neighbour; ours sums neighbours equally.
Training: the guide is an EMA at epoch 2,000 [15]; ours stop at 120.
Inputs: the oracle also reads nuclear-charge features.

Why we keep ours on our FM
Ours is disjoint **by construction**; TFG's only **by inference** — its saved arguments are overwritten by its own training script, and the oracle's training IDs are unknown. The price is a wider band (ours / TFG δ : 1.12× μ , 3.07× α , 2.00× gap), which is exactly why in-band never crosses backends.

Disjointness test (μ , 3,000 molecules): guide wins by >2× on 30.9%, oracle on 40.9%, neither on 28.2% — against 37.4 / 37.4 / 25.3% predicted for disjoint halves. Replicated on α and gap.

The data split, and why it has this shape

Split	Trains or supplies	n
train_a	generator (FM and VP) and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	never trained or selected on	13,083

δ calibration: 3,000 molecules from train_a + train_b; each of our nets is scored only on the half it did not train on.

QM9 [1], 133,885 molecules, pinned by SHA-256; split fixed by seed 20260917.

- 1 · f_A and f_B never share data**

Guidance is an optimiser aimed at f_A; pushed hard it finds inputs that fool it. The checker needs different blind spots, so the guide-evaluator gap stays a usable reward-hacking signal.
- 2 · The generator may share with f_A**

It judges nothing. This keeps the claim to one sentence: **f_B never saw a molecule the generator or the guide trained on.**
- 3 · Halves, as the literature does**

EDM's conditional protocol and TFG's EDMsecond train on one ~50K half; TFG shipped a full-data EDMfull and *chose* EDMsecond [2, 15]. Using both halves would set twice the data beside every number we compare to. 70/30 was rejected for the same reason.

The cost we accept: half the data. A published half-data EDM loses only ~0.3 atom-stability points [2, 25] — so the split does not explain our gap in 2a.

The guidance arms, side by side

Shared frame. $m = x_t + (1 - t)v\theta \rightarrow$ frozen $f_A \rightarrow$ pull back through $J = \partial m / \partial x_t \rightarrow$ clip $\rightarrow$ add to the velocity ($t \geq 0.5, w = 1$).

Arm	Correction	What it changes	Passes per cell (2e)
DPS-style plug-in [12]	$G = \frac{y - f_A(m)}{s^2} J^\top \nabla f_A(m)$	the property at the endpoint; the baseline	1,000 / 400
TMPD-inspired [22]	$G = \frac{y - f_A(m)}{s^2 + g^\top \Sigma g} J^\top g, \quad \Sigma = k J, \quad g = \nabla f_A(m)$	shrinks the step where the endpoint is uncertain	1,200 / 400
LGD-MC [23]	$G = J^\top \sum_k \omega_k \frac{y - f_A(m + rz_k)}{s^2} \nabla f_A(m + rz_k), \quad r^2 = \frac{1}{d} \text{tr } \Sigma$	averages the likelihood over K draws around m	1,200 / 2,000
TFG [15]	$L(x) = \log \frac{1}{K} \sum_k \exp \left[- \left(\frac{f_A(x + \epsilon_k) - y}{\text{MAD}} \right)^2 \right]$ $\Delta_t = \rho \nabla_{x_t} L(m), \quad \Delta_0 : m \leftarrow m + \mu \nabla_m L(m) \quad (4 \text{ steps, no } J)$	adds steps on the predicted molecule itself	800 / 2,000
BDG (ours)	$\text{num}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}), \quad w_{\text{eff}} = 1 + \eta e$	plug-in with a servoed deviation gain	1,000 / 400

BDG in the sampler: Algorithm 1

Algorithm 1 One BDG-guided Euler step

red: BDG's only change to plug-in

Require: batch x_t of size B ; frozen v_θ and f_A ; target y ; scale $s > 0$; gain $\eta \geq 0$; setpoint $\tau > 0$; strength w ; step h ; start $t_{\min}$

- 1: $v \leftarrow v_\theta(x_t, t; M)$
- 2: **if** $t < t_{\min}$ **return** $\Pi(x_t + h v)$ ▷ guide only for $t \geq t_{\min}$
- 3: $m \leftarrow x_t + (1 - t) v_\theta(x_t, t; M)$, with gradients enabled
- 4: $F_i \leftarrow f_A(m_i)$; $\bar{F} \leftarrow B^{-1} \sum_i F_i$
- 5: $V_b \leftarrow (B - 1)^{-1} \sum_i (F_i - \bar{F})^2$ if $B > 1$, else τ^2
- 6: $e \leftarrow (V_b - \tau^2) / \tau^2$
- 7: $a_i \leftarrow \text{stopgrad} \{ [(y - F_i) - \eta e (F_i - \bar{F})] / s^2 \}$ ▷ coefficients detached
- 8: $G \leftarrow \text{VJP}_{x_t}(F, a)$ ▷ one backward pass, as plug-in
- 9: $C_i \leftarrow w (1 - t) / \max(t, \varepsilon) G_i$
- 10: $C_i \leftarrow C_i \min(1, \|v_i\| / \max(\|C_i\|, \varepsilon))$ ▷ velocity-relative cap
- 11: **return** $\Pi\{x_t + h(v + C)\}$ ▷ Π : mask and centre

Checked in code

$\eta = 0$ gives plug-in's field bit for bit; the reduction and the bound hold numerically.

Cost

Plug-in's network calls; the red lines are scalar batch reductions.

The batch is the unit

$B = 500$, so each 2,000-sample cell runs four independent controllers.

On DNA only Π changes: clamp at zero and renormalise each position; decode by argmax at the end (3a).